

Technical Documentation: Competitor List Building Workflow

1. Overview

Workflow Name: Competitor List Building Agent **Objective:** To automate the extraction of job posting data from competitors, clean and format the raw data, and store it in a structured Google Sheet for analysis. **Primary Tools:** n8n, Google Jobs (via SERP API), JavaScript (n8n Code Node), Google Sheets.

2. Prerequisites

Before constructing this workflow, ensure you have the following credentials and setups ready:

n8n Instance: Active access to n8n (Cloud or Self-hosted).

Google Workspace Account: An authenticated connection in n8n for Google Sheets (OAuth2 recommended).

Search API Credentials: An API key for a SERP API provider (e.g., SerpApi) to perform Google Jobs searches.

Target Google Sheet: A prepared Google Sheet with the following column headers in exactly this order:

company_name

website

job_position

job_requirements

location

3. Architecture

The workflow follows a standard ETL (Extract, Transform, Load) pattern:

Trigger (Manual/Schedule): Initiates the workflow.

Extract (Google Jobs Search): Fetches raw job posting data via API.

Transform (Code Node): Parses the JSON payload and maps specific fields.

Load (Google Sheets): Appends the cleaned data into the prepared spreadsheet.

4. Step-by-Step Construction Guide

Step 1: The Trigger Node

Add a **Manual Trigger** node to the canvas.

Note: Use this node during development and testing. Once the workflow is verified, replace it with a **Schedule Trigger** node to run daily or weekly.

Step 2: The Data Extraction Node (Google Jobs Search)

Add an **HTTP Request** node (or a dedicated SERP API node if available in your n8n version).

Rename the node to Google Jobs search.

Connect the output of the Trigger node to this node.

Configuration:

Method: GET

URL: Enter the endpoint for your SERP API provider.

Authentication: Add your API Key.

Query Parameters: Set the search string (e.g., "Tayana Solutions" jobs).

Execute the node to verify it returns a successful JSON response containing the job postings.

Step 3: The Data Transformation Node (Code Node)

Add a **Code** node and rename it to Code in JavaScript.

Connect the output of the Google Jobs search node to this node.

Configuration:

Mode: Run Once for All Items (or Run Once for Each Item depending on your mapping preference).

Language: JavaScript

Script implementation: Write a script to map the messy API response to your exact Google Sheet headers. (*Example Script Structure*):

```
// Target headers: company_name, website, job_position,
job_requirements, location
const rawData = $input.all()[0].json.jobs_results; // Adjust path based
on API provider
const cleanData = [];

for (const job of rawData) {
  cleanData.push({
    json: {
      company_name: job.company_name || "N/A",
      website: job.related_links ? job.related_links[0].link : "N/A",
      job_position: job.title || "N/A",
      job_requirements: job.description || "N/A",
      location: job.location || "N/A"
    }
  });
}
return cleanData;
```

Execute the node to ensure the output is a clean array of objects with only the required keys.

Step 4: The Data Storage Node (Google Sheets)

Add a **Google Sheets** node and connect it to the Code in JavaScript node.

Rename the node to Append row in sheet.

Configuration:

Credential for Google Sheets: Select your authenticated account.

Resource: Sheet

Operation: Append Row

Spreadsheet: Select your target document from the dropdown (e.g., "Competitor List").

Sheet: Select the specific tab (e.g., "Sheet1").

Data Mode: Map Automatically (since the JS node keys perfectly match the sheet headers).

Execute the node to push a test row into your live Google Sheet.

5. Testing and Deployment

Verify the Google Sheet to ensure all columns populated correctly without formatting errors.

Check the n8n execution logs for any warnings or errors.

Delete the **Manual Trigger** and connect a **Schedule Trigger** (e.g., set to run every Monday at 9:00 AM).

Toggle the workflow status in the top right corner of the n8n canvas to **Active**.